DETAILED DESCRIPTION OF THE INVENTION

Device Architecture Overview

Referring generally to the embodiments described herein, the penile geometry profiling system comprises a scanning device configured to acquire spatial distance data and longitudinal positional data relative to a penile anatomical structure. In various embodiments, the device includes a housing structure defining a reference frame from which radial distance measurements are obtained. The housing may be handheld, partially enclosing, C-shaped, open-frame, linear, or otherwise configured to maintain a controlled spatial relationship relative to the anatomical surface during measurement.

At least one time-of-flight (ToF) ranging sensor is positioned within the housing and oriented toward the anatomical surface. The ToF sensor is configured to emit a signal and detect reflected return signals to determine radial distance between the sensor reference plane and the anatomical surface. In certain embodiments, multiple ToF sensors may be positioned circumferentially or at offset angles to improve surface sampling resolution. The ranging subsystem may operate using direct time-of-flight, indirect time-of-flight, phase-shift, pulsed, or structured light techniques.

An optical motion sensing component is further incorporated into the device and configured to detect longitudinal displacement of the device relative to the anatomical surface during scanning. The optical motion sensing component may include an optical flow sensor, speckle-based displacement detector, emitter-receiver tracking pair, or other surface-relative motion detection system. Motion data may be sampled continuously or at defined intervals and correlated temporally with distance measurements.

In some embodiments, the device further includes an inertial measurement unit (IMU) configured to detect angular orientation, tilt, or acceleration. IMU data may be used to compensate for unintended user movement or angular deviation during scanning to preserve reconstruction accuracy.

The device may further comprise a processing unit configured to synchronize distance measurements with motion data, timestamp collected samples, and transmit acquired datasets to an onboard or remote reconstruction engine. Power may be supplied by an internal rechargeable battery, removable power source, or external connection. Communication with external devices may occur through wired or wireless protocols including but not limited to Bluetooth, Wi-Fi, or other digital communication standards.

The architecture described herein is not limited to any specific housing geometry, sensor placement, or electronic configuration and encompasses variations capable of acquiring radial distance measurements and correlated longitudinal displacement data sufficient to reconstruct a user-specific penile anatomical geometry profile.